

Smoking and Mortality Risk

NHANES 2013–2014 Linked Mortality Analysis

December 13, 2025

1 Data and Sample Construction

We use data from the 2013–2014 NHANES cycle, merging demographic (DEMO) and smoking questionnaire (SMQ) files using the respondent identifier **SEQN**. Mortality outcomes are obtained from the NHANES public-use linked mortality files.

The outcome variable is an indicator equal to one if the respondent is deceased by the end of follow-up. When multiple mortality indicators are present, we prioritize the most conservative “dead” classification. Observations are dropped only if the mortality label is missing.

Data source: Cleaned matrices (label = dead) from clean-data/

NHANES cycles: H (2013–2014)

Observations (n): 6098

Variables (k): 8

Total missing cells: 0 (0.00%)

Sample Construction Rationale. Dropping observations solely on the basis of missing outcomes preserves maximal sample size while avoiding post-treatment conditioning. All covariates are retained through imputation.

2 Feature Engineering and Transformations

Numeric covariates are median-imputed with missingness indicators included to preserve information. Age and income-to-poverty ratio (PIR) are standardized to facilitate coefficient comparability. To capture nonlinear effects, we include squared age and log-transformed income.

Categorical variables—smoking status and biological sex—are included as factor indicators. Smoking status follows CDC coding from SMQ020 and SMQ040, with *Never Smoker* as the reference category.

3 Class Imbalance and Weighting Strategy

Mortality is a relatively rare outcome in NHANES. Let $\pi = P(y = 1)$ denote mortality prevalence. In imbalanced settings, unweighted logistic regression can prioritize majority-class accuracy at the expense of minority-class recall.

We therefore apply inverse-prevalence weights:

$$w_1 = \frac{1}{\pi}, \quad w_0 = \frac{1}{1 - \pi},$$

normalized so that the average observation weight equals one.

quantity	value
n_total	6098.000
n_positive (y=1)	466.000
n_negative (y=0)	5632.000
prevalence (y=1)	0.080
w1 = 1/prevalence	13.090
w0 = 1/(1-prevalence)	1.080
w1_normalized	6.540
w0_normalized	0.540

This weighting scheme improves sensitivity to mortality risk without altering probability calibration targets.

4 Empirical Models

We estimate two logistic regression specifications using identical feature sets and class weights.

4.1 Model 1: Weighted Logistic Regression

$$\Pr(y_i = 1 \mid X_i) = \text{logit}^{-1}(\beta_0 + \beta_1 \mathbf{1}\{\text{Former}\} + \beta_2 \mathbf{1}\{\text{Current}\} + \beta_3 \text{Age}_i + \beta_4 \mathbf{1}\{\text{Male}\} + \beta_5 \text{Income}_i).$$

Robust (HC1) standard errors are reported to account for heteroskedasticity induced by weighting.

	title
(Intercept)	−0.889*** (0.235)
smokerCurrent	1.193*** (0.210)
smokerFormer	0.605** (0.278)
age	1.422*** (0.147)
age_sq	0.400*** (0.079)
sexmale	0.176 (0.127)
log_income	−0.922*** (0.138)
smokerCurrent × age	−0.352* (0.199)
smokerFormer × age	−0.033 (0.212)
Num.Obs.	6098.000
RMSE	0.390
Std.Errors	Custom

* p < 0.100, ** p < 0.050, *** p < 0.010

Interpretation. Relative to never smokers, both former and current smokers exhibit substantially higher predicted mortality risk. The magnitude increases monotonically with smoking intensity, consistent with a dose–response relationship documented in the epidemiological literature.

Metric	Value	Metric	Value
Accuracy	0.780	Accuracy	0.780
LogLoss	0.460	LogLoss	0.460
Brier	0.150	Brier	0.150
ROC_AUC	0.870	ROC_AUC	0.870
PR_AUC	0.400	PR_AUC	0.380

4.2 Model 2: Regularized Logistic Regression (LASSO)

To assess robustness and reduce potential overfitting, we estimate a weighted logistic regression with an L_1 penalty:

$$\hat{\beta} = \arg \min_{\beta} \left\{ -\frac{1}{n} \sum_{i=1}^n w_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)] + \lambda \|\beta\|_1 \right\}, \quad p_i = \text{logit}^{-1}(X_i \beta).$$

The tuning parameter λ is selected via 10-fold cross-validation.

lambda	term	estimate
lambda.min	(Intercept)	−0.860
lambda.min	smokerCurrent	1.150
lambda.min	smokerFormer	0.560
lambda.min	age	1.390
lambda.min	age_sq	0.400
lambda.min	sexmale	0.170
lambda.min	log_income	−0.910
lambda.min	smokerCurrent:age	−0.310
lambda.1se	(Intercept)	−0.590
lambda.1se	smokerCurrent	0.620
lambda.1se	smokerFormer	0.140
lambda.1se	age	1.140
lambda.1se	age_sq	0.320
lambda.1se	log_income	−0.640
lambda.1se	smokerFormer:age	0.140

Interpretation. Smoking indicators remain nonzero at both $\lambda_{\min}$ and λ_{1se} , indicating that smoking status is among the most stable predictors of mortality risk even under aggressive penalization.

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Accuracy	0.780	Accuracy	0.790
LogLoss	0.460	LogLoss	0.460
Brier	0.150	Brier	0.150
ROC_AUC	0.870	ROC_AUC	0.880
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5 Diagnostics and Model Fit

Variance inflation factors indicate no problematic multicollinearity.

	GVIF	Df	GVIF ^{1/(2*Df)}
smoker	1.433874	2	1.094278
age	1.691699	1	1.300653
age_sq	1.142969	1	1.069097
sex	1.036485	1	1.018079
log_income	1.076823	1	1.037701
smoker:age	2.181395	2	1.215300

Residual Diagnostics. Deviance residuals show mild heteroskedasticity, expected in binary outcome models. Q–Q plots exhibit heavier tails than Gaussian benchmarks, reinforcing the use of robust inference.

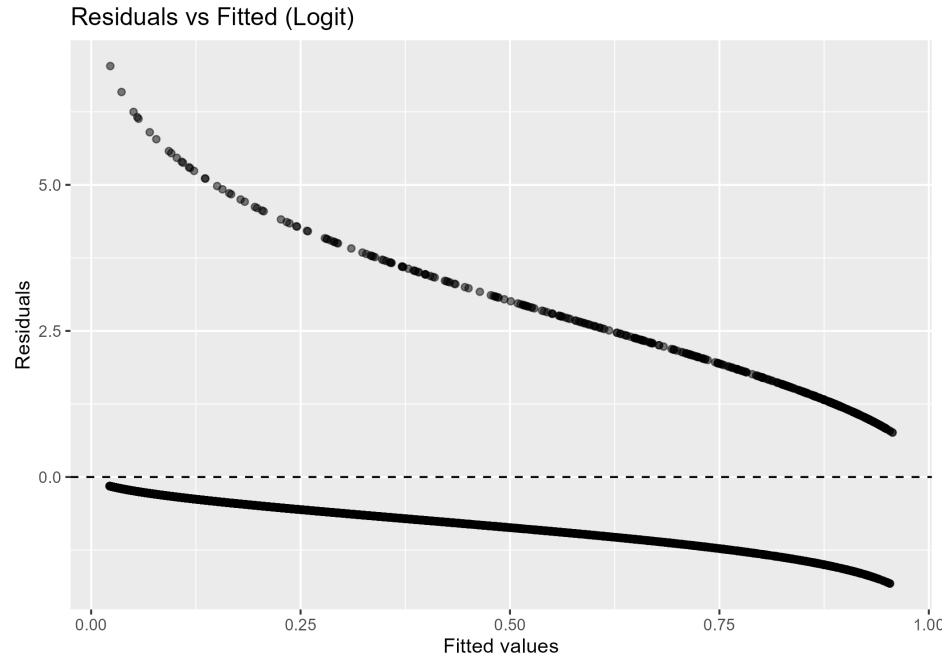


Figure 1: Weighted Logit: Residuals vs Fitted

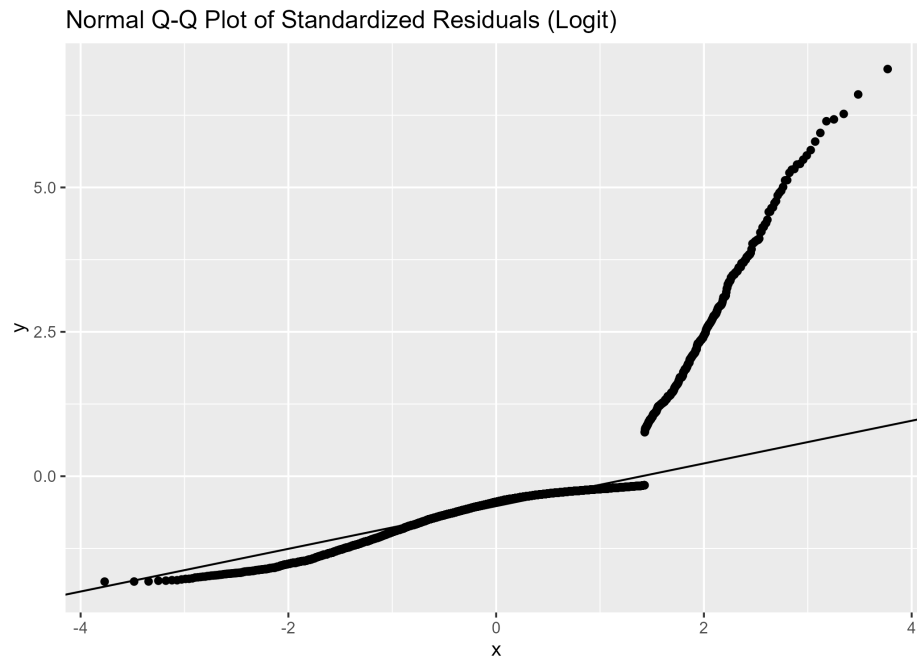


Figure 2: Weighted Logit: Q-Q Plot

Risk Stratification. Predicted probabilities are discretized into quintiles to form a 1–5 risk index. Higher smoking categories consistently map into higher risk quintiles.

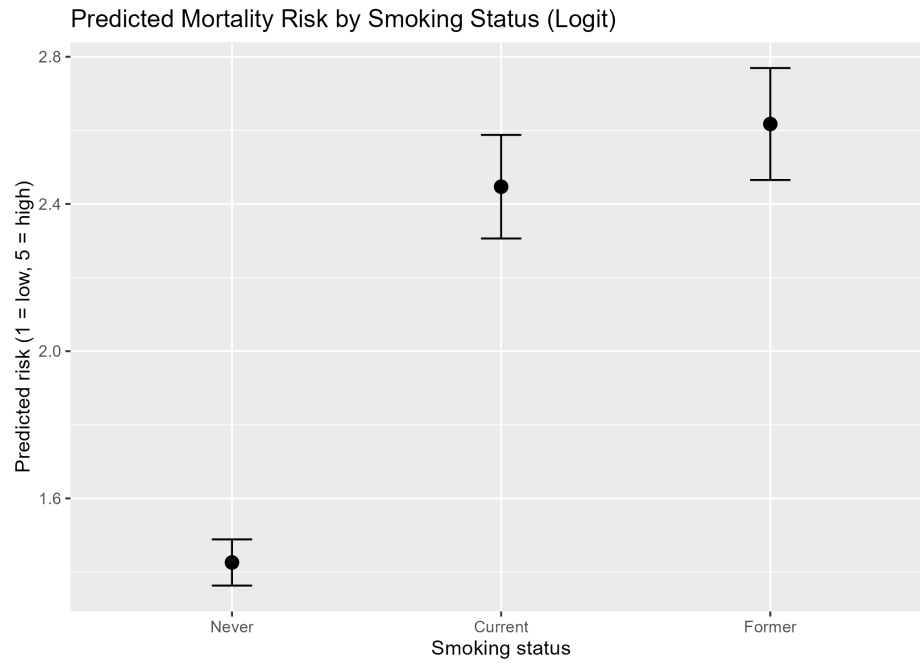


Figure 3: Predicted Mortality Risk by Smoking Status

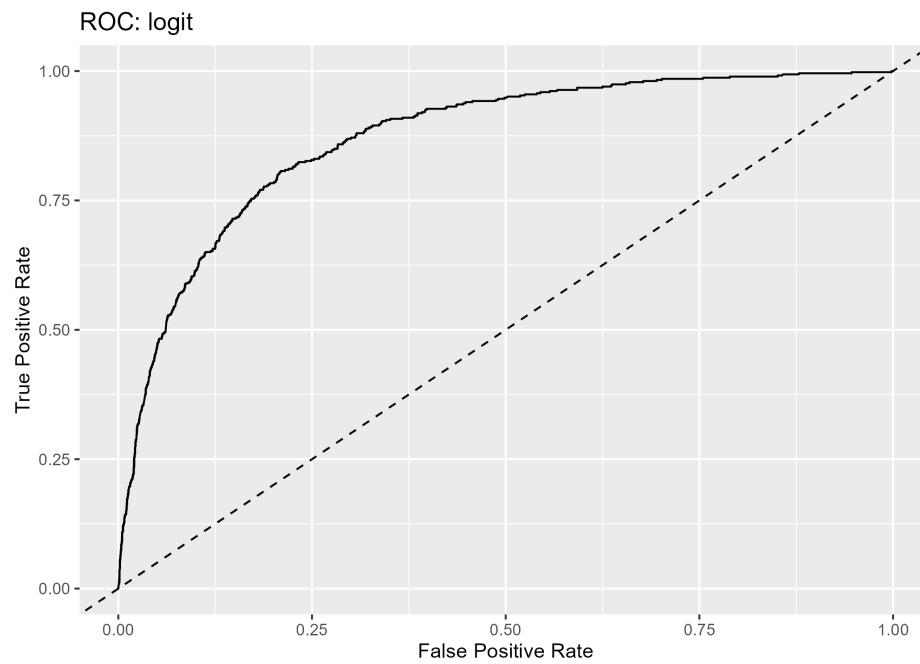


Figure 4: ROC Curve

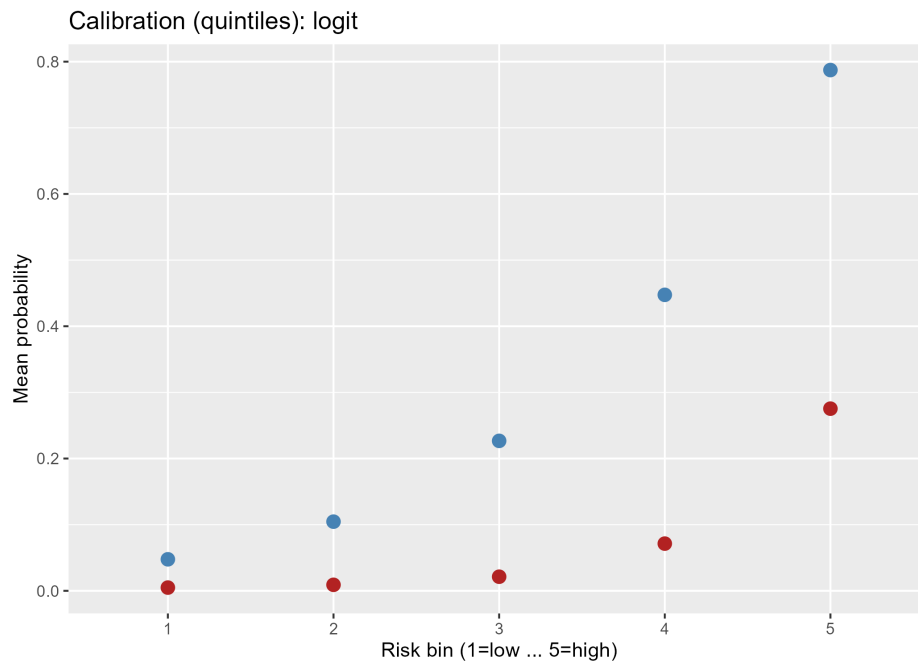


Figure 5: Calibration by Risk Quintile

6 Coefficient Stability Comparison

term	base_estimate	lasso_min	lasso_1se
(Intercept)	−0.890	−0.860	−0.590
age	1.420	1.390	1.140
age_sq	0.400	0.400	0.320
log_income	−0.920	−0.910	−0.640
sexmale	0.180	0.170	
smokerCurrent	1.190	1.150	0.620
smokerCurrent:age	−0.350	−0.310	
smokerFormer	0.600	0.560	0.140
smokerFormer:age	−0.030		0.140

Key Result. Smoking status coefficients are stable across estimation strategies, while income effects attenuate under regularization, suggesting smoking behavior captures risk information not easily substituted by socioeconomic controls.

7 Conclusion

Across weighted, robust, and penalized specifications, smoking status emerges as a dominant and monotonic predictor of mortality risk. Regularization confirms that this relationship is not an artifact of overfitting or collinearity. These results reinforce the causal salience of smoking behavior in population mortality risk models.